

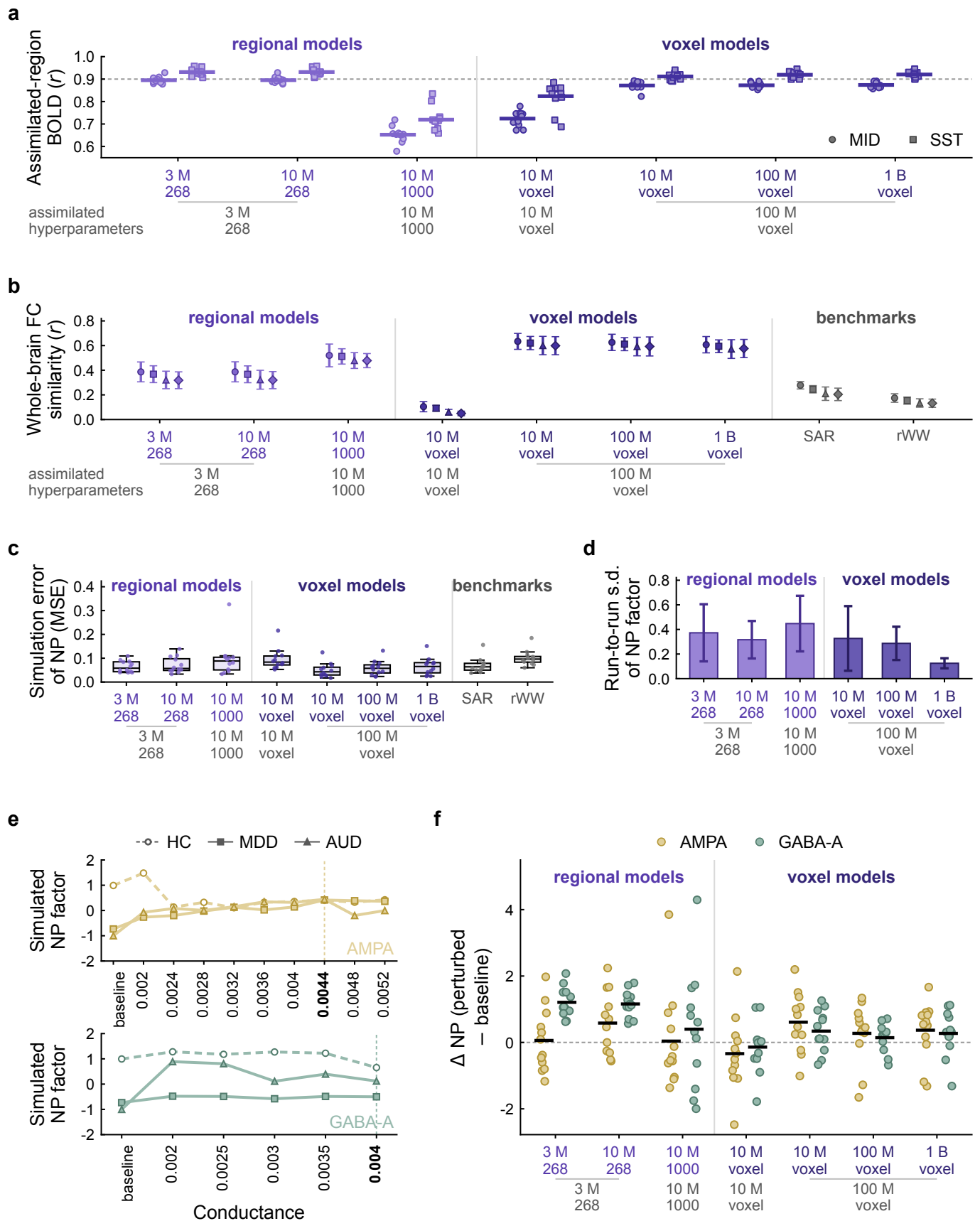


Fig. 3 | Multiscale validation of the digital twin brain. Seven DTB builds are compared throughout; they differ in the number of simulated neurons, the neuron count of the assimilated hyperparameter set and the spatial resolution of the simulation target, so every panel carries a second tick row naming the hyperparameter set, labelled once per set with a rule spanning the builds that share it. Colour encodes model family (regional versus voxel) only; benchmarks (SAR, rWW) are grey. **a**, Pearson correlation between simulated and measured BOLD within the assimilated regions (MID, 51 Shen labels; SST, 47), averaged over assimilated voxels; one subject $\times$ task session, 24 points per build, bar, group mean; two-sided paired t tests, $n = 24$, uncorrected. **b**, whole-brain FC similarity, mean $\pm$ s.d. across the 12 twins of the correlation between simulated and measured FC (23,436 edges among 217 nodes), per task condition (marker shape); $n = 12$ per point; descriptive. **c**, simulation error of the 12-edge NP profile against one empirical reference (FC recomputed from the simulated voxels); median, 25th–75th percentiles, $1.5 \times$ IQR whiskers, all 12 twins overlaid; paired t tests, $df = 11$, uncorrected. Benchmarks are scored against the Shen-268 task FC their coupling was fitted to and are not on the DTB scale. **d**, run-to-run variability: mean across twins of the within-subject s.d. of the NP factor over 5 repeats; error bars, s.d. across twins; the single-run 10 M voxel own-hyperparameter-build is omitted; descriptive. **e**, NP factor across the AMPA (upper; 9 settings, 0.0020–0.0052) and GABA-A (lower; 5 settings, 0.0020–0.0040) conductance sweeps in three twins (HC, MDD, AUD), stacked. The first column of each segment is that twin's own unperturbed baseline, the same value in both segments; the GABA-A settings themselves are applied on top of the AMPA perturbation, as in **f**. Colour marks the perturbation, marker and dash pattern the twin; vertical dashed line and bold tick, AMPA 0.0044 and GABA-A 0.0040; descriptive only, no test at $n = 3$. **f**, change in summed NP after AMPA and after GABA-A perturbation in all seven builds; points, twins; bar, group mean; $n = 12$ per build $\times$ perturbation; two-sided one-sample t tests against zero, $df = 11$, Benjamini–Hochberg FDR across the 14 tests. Only the two 268-region builds survive correction, and only under GABA-A; GABA-A is applied on top of AMPA.